CTI–Aware Malicious URL Detection

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DATA MINING AND MACHINE LEARNING (24CY731)

Introduction

- Malicious URLs pose a serious and growing threat, enabling **phishing attacks, malware distribution,** and website defacement
- Traditional rule-based detection methods struggle to keep up with **sophisticated, evasive threats,** often failing to detect newly generated, obfuscated, or mimicking malicious URLs
- Cybercriminals use advanced **evasion techniques** such as **encoding, obfuscation,** and **URL manipulation** to bypass simple blacklist and rule-based filters
- There is an **increasing need for intelligent, Machine Learning-based detection systems** that can adapt to evolving threats and accurately detect malicious URLs
- **Incorporating Cyber Threat Intelligence features** and **analyzing false negatives** are critical to improving the accuracy and robustness of such detection systems
- Detecting malicious URLs not only protects users from phishing, data theft, and malware infections but also prevents financial losses and protects organizational networks

PROBLEM STATEMENT

To **develop** an **intelligent machine learning**-based system for detecting malicious URLs by integrating **Cyber Threat Intelligence (CTI)**, addressing the limitations of traditional **rule-based methods** against sophisticated evasion techniques, and evaluating model robustness against adversarial attacks.

PROJECT OBJECTIVES

- **Detect malicious URLs using Machine Learning** models with high accuracy and reliability.
- **Improve detection performance by integrating Cyber Threat Intelligence (CTI)**-inspired features such as risk scores, domain reputation, and historical threat data.
- **Analyze false negatives** to understand misclassification patterns and identify model weaknesses.
- **Test adversarial robustness** by generating and evaluating URLs designed to evade detection through encoding, obfuscation, and other evasion techniques.

This project not only focuses on building an effective malicious URL detection system but also emphasizes the importance of **analyzing model weaknesses and improving robustness to real-world threats through CTI integration and adversarial evaluation.**

Malicious URL Dataset Preparation

Original Dataset
651,191 URLs
4 Classes

→ **Invalid Domains Removed**
- 17 URLs

→ **Duplicate Domains Removed**
- 456,554 Domains

→ **Final Dataset**
194,620 Domains

Class Distribution

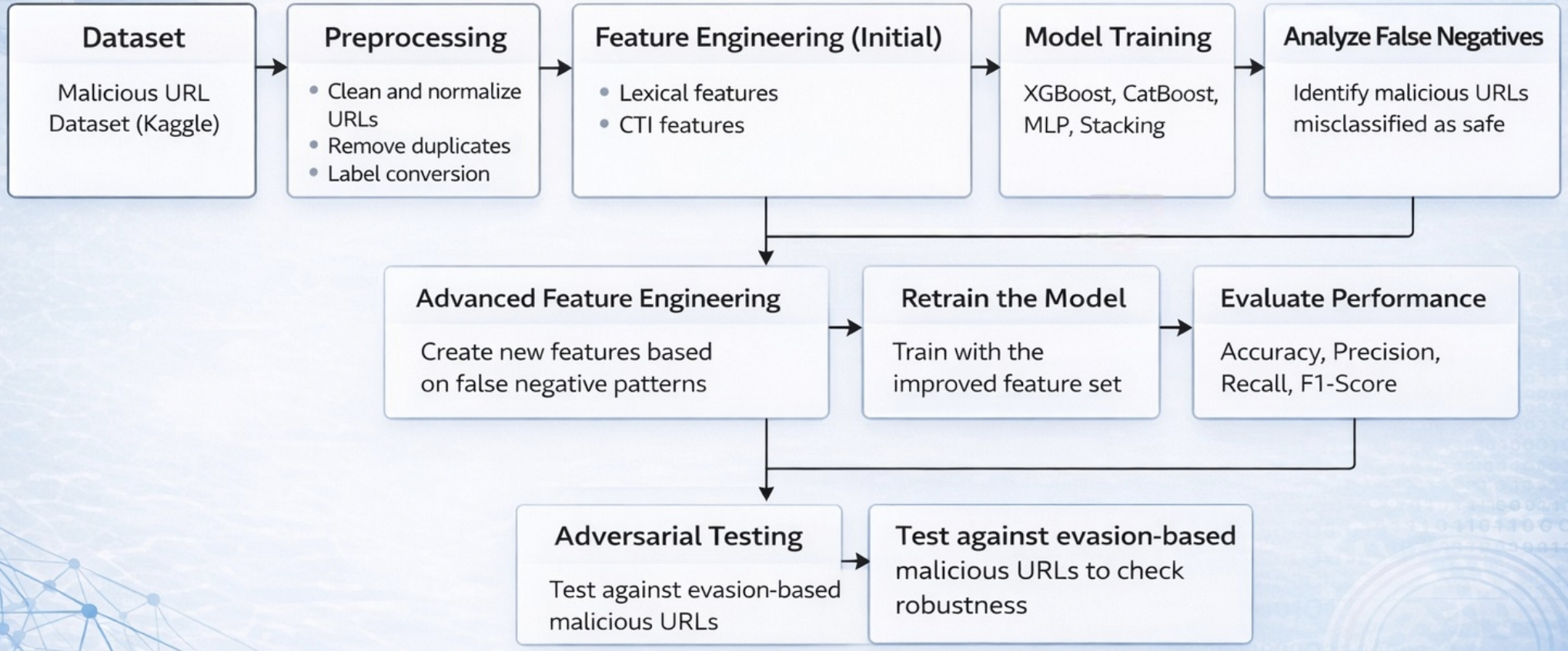
Dataset Overview

- Total Domains: 194,620
- Benign: 131,108
- Malicious: 63,512
- Classes: Binary (0 = Benign, 1 = Malicious)

Sample Domains

- br-icloud.com.br
- mp3raid.com
- garage-pirenne.be
- adventure-nicaragua.net

SYSTEM WORKFLOW



Feature Engineering (Phase 1)

CTI FEATURES

- **Risk Score:** Sum of CTI-flags indicating suspicious activity
- **suspicious_tld_flag:** TLD is frequently abused (e.g.,xyz, tk)
- **suspicious_keyword_flag:** Contains keywords like "login", "secure", "update"
- **length_anomaly:** Domain length > 25 characters
- **Historical and Domain History:** Domain age, reputation, and past threats

LEXICAL FEATURES

- Length
- Digit Count
- Special Char Count
- Entropy
- Bigram Log Probability

IMPORTANT FEATURES

- Entropy
- Digit Ratio
- **Risk Score:** CTI-based flag aggregation

MODELS USED (PHASE 1)

- Random Forest
- XGBoost
- CatBoost
- Multi-Layer Perceptron (MLP)
- CNN

INITIAL RESULTS

- **Stacking Model** (XGB + CB + MLP → XGBoost)
- **Best Accuracy: 90.4%**
- **Best ROC-AUC: 0.966**

Model	Accuracy	ROC-AUC	Recall	F1 Score
Stacked Model	0.904	0.966	0.84	0.85
Deep MLP	**0.904**	0.965	0.82	0.85
XGBoost	0.902	0.963	0.82	0.85
CatBoost	0.901	0.962	0.82	0.84
CNN	0.894	0.962	0.84	0.89
Random Forest	0.898	0.958	0.74	0.89

Stacking Model
Meta: XGBoost

Base: XGB + CB + MLP

Phase 1 Summary: CTI + Lexical features with ensemble models achieved **90.4% Accuracy** and **0.966 ROC-AUC.**

False Negative Analysis

False Negatives Overview

- **False Negatives** Analyzed: 2,284
- Average Risk Score: 0

Problem Identified

CTI Features Not Enough

CTI-based features alone did not flag low-risk, stealthy attacks.

Model Missing Stealth Attacks

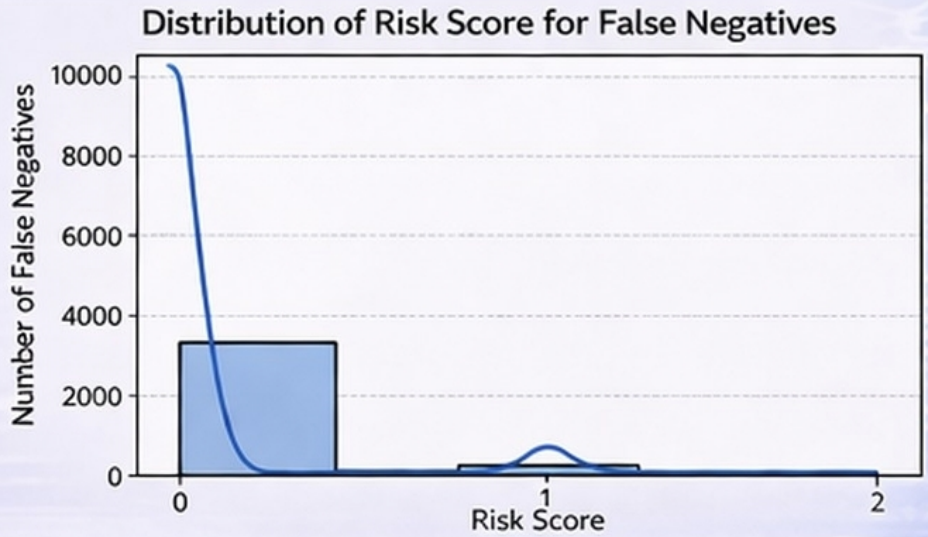
Model struggled to detect stealthy and obfuscated malicious URLs.

Key Insight

Attackers Mimic Benign URLs

Attackers disguise malicious URLs as benign ones to evade detection.

- Use trusted domains (e.g., googlesecurity-webs[].com)
- Add common keywords (e.g., account, secure)
- Employ URL shorteners to hide full links



FALSE NEGATIVE ANALYSIS: CTI-based features alone did not flag low-risk, stealthy attacks. Model struggled to detect stealthy and obfuscated malicious URLs.

Phase 2 Improvements and Final Models

Advanced Cyber Threat Intelligence (CTI) Features Used:

- **Suspicious TLD Flag:** Identifies abusive top-level domains like .xyz, tk.
- **Suspicious Keyword Flag:** Detects keywords like "login", "secure", "account"
- **Length Anomaly:** Flags domains with unusually long lengths.
- **Contains IP Address:** Identifies domains using IP addresses. directly.
- **Brand Flag:** Detects impersonation of popular brands.
- **Typosquatting Score:** Measures similarity to famous brand domains.
- **Risk Score:** Aggregates CTI features into a single risk metric.

Superior Performance Achieved:

- **Stacked Model:** Combined XGBoost, CatBoost, and Deep MLP models.
- **Best Accuracy:** Achieved a 94.0% accuracy rate.
- **Robust Detection:** Outperformed individual models in detecting malicious URLs.

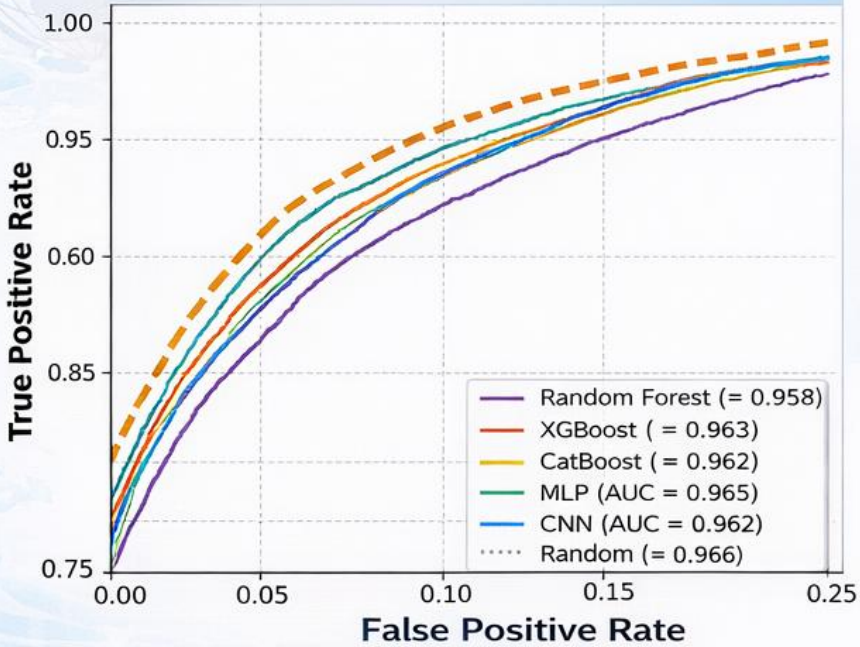
Model Comparison (Phase 2)

Model	Accuracy	ROC-AUC	Recall	F1 Score
Stacked Model	0.9404	0.9831	0.88	0.94
Deep MLP	0.9381	0.9816	0.87	0.94
XGBoost	0.9326	0.9785	0.86	0.93
CatBoost	0.9276	0.9764	0.87	0.93

The Stacked Model demonstrated superior performance on the original test set compared to individual models. it exhibited better detection capabilities overall.

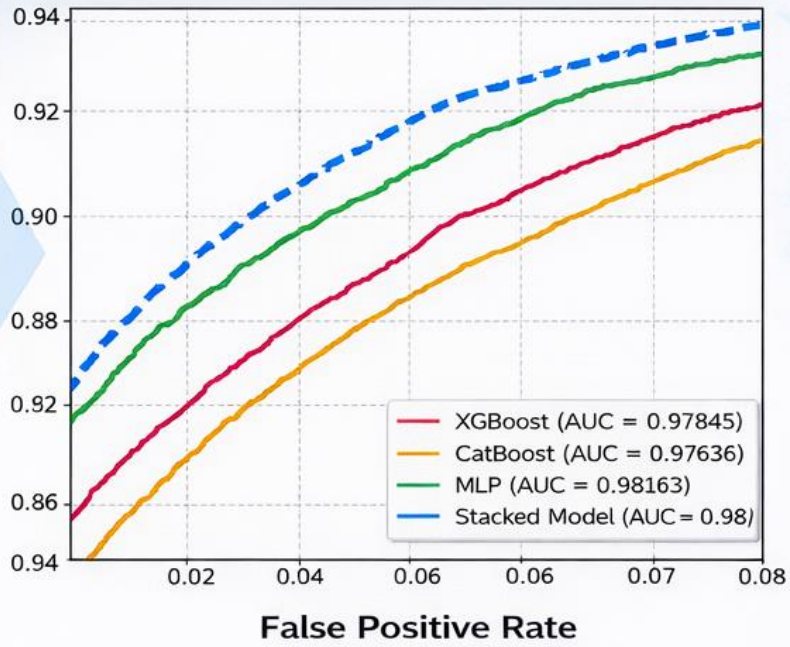
ROC Curve Comparison

Phase 1 ROC Curves



Advanced Ensemble Stacking in Phase 2

Phase 2 ROC Curves



Key Insights

- Phase 2 featured a focused selection of top-performing models compared to Phase 1
- The Stacked Model in Phase 2 significantly improves, **yielding the highest AUC of 0.983**
- All models in Phase 2 show improved ROC curves, indicating **better performance**
- Phase 2 models consistently achieve higher **true positive** rates and lower **false positive rates**

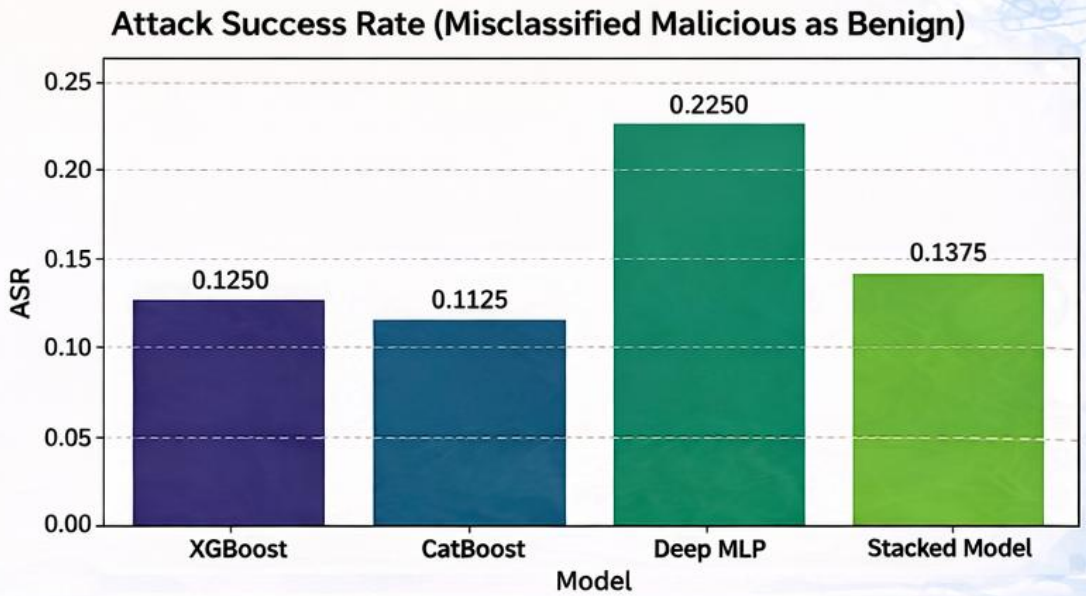
Adversarial Testing & Evaluation

Adversarial Testing Setup

- Generated **110 adversarial URLs**
- Simulated **real-world evasion techniques**
- Goal: Evaluate **model robustness** against attacks

Attack Types:

- **Encoding attacks** (URL encoding, double encoding)
- **Homograph attacks** (lookalike domains)
- **Null byte** injection
- **Query parameter** obfuscation
- **Domain/IP-based** manipulation



Evaluation Metrics

Attack Success Rate (ASR)

- Percentage of malicious URLs misclassified as benign

Lower ASR – better robustness

Accuracy Drop

- Measures performance degradation under attack

Results Summary (ASR)

Model	ASR (↓ Better)	Robustness
CatBoost	0.1125	Most Robust
XGBoost	0.1250	Moderate
Stacked Model	0.1375	Balanced
Deep MLP	0.2250	Most Vulnerable

Final Takeaway

"Models performing well on clean data are still vulnerable to adversarial attacks."

Adversarial testing helps evaluate real-world reliability of ML-based security systems.

Conclusion & Future Work

Conclusion

- The proposed system demonstrates that integrating **Cyber Threat intelligence (CTI)** with **machine learning** significantly enhances malicious URL detection performance and adaptability.
- Hybrid feature engineering (lexical + **CTI-based features)** improves **detection of obfuscated and evolving threats**.
- **False negative analysis** reveals key evasion patterns, helping refine detection strategies.
- **Adversarial testing** highlights model yulnerabitities, emphaszing **techniques such as URL obfuscation and feature manipulation**, emphasizing the need for **robust and resilient detection** mechanisms.

Future Work

- Enable **real-time CTI integration** using **live threat feeds** (e.g., phishing databases, blacklist APIs).
- Explore **deep learning and transformer-based architectures** for advanced URL representation learning.
- Implement **adversarial training** and **defense techniques** to improve robustness against evasion attacks.
- Incorporate **zero-day threat samples** and **dynamic threat intelligence feeds** to enhance generalization.
- Extend the system towards a **real-world deployment pipeline** (API / SOC integration).